

Hyeongseop Rha

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RESEARCH INTERESTS

Human-Centric AI Communication: Building AI that perceives human nuance and truly communicates by **integrating Multi-Modal LLMs, Multimodal Emotion Reasoning, and Visual Speech Recognition.**

EDUCATION

KAIST

Ph.D. Candidate in Electrical Engineering

Mar. 2023 – Present
Daejeon, Korea

- Advisor: Prof. Yong Man Ro
- Integrated Vision Language Lab (IVLLab)

Yonsei University

B.S. in Electrical and Electronic Engineering

Feb. 2017 – Feb. 2023
Seoul, Korea

PUBLICATIONS

(* denotes equal contribution)

2026

Decoding Strategies for Diffusion-Based ASR: A Systematic Evaluation of Confidence-Based Thresholding

- Jeonghun Yeo, Minsu Kim, **Hyeongseop Rha**, and Yong Man Ro
- Conference on Empirical Methods in Natural Language Processing (**EMNLP**), 2026

Diffusion Large Language Models for Visual Speech Recognition

- Jeonghun Yeo, Chae Won Kim, **Hyeongseop Rha**, and Yong Man Ro
- Conference on Empirical Methods in Natural Language Processing (**EMNLP**), 2026

Emotion-Coherent Reasoning for Multimodal LLMs via Emotional Rationale Verifier

- **Hyeongseop Rha**, Jeonghun Yeo, Yeonju Kim, and Yong Man Ro
- The Association for the Advancement of Artificial Intelligence (**AAAI**), 2026

2025

MMS-LLaMA: Efficient LLM-based Audio-Visual Speech Recognition with Minimal Multimodal Speech Tokens

- Jeonghun Yeo*, **Hyeongseop Rha***, Se Jin Park, and Yong Man Ro
- Proceedings of the Annual Meeting of the Association for Computational Linguistics (**ACL Findings**), 2025

Personalized Lip Reading: Adapting to Your Unique Lip Movements with Vision and Language

- Jeonghun Yeo, Chae Won Kim, Hyunjun Kim, **Hyeongseop Rha**, Seunghee Han, Wen-Huang Cheng, and Yong Man Ro
- The Association for the Advancement of Artificial Intelligence (**AAAI**), 2025

Learning What to Attend First: Modality-Importance-Guided Reasoning for Reliable Multimodal Emotion Understanding

- **Hyeongseop Rha**, Jeonghun Yeo, Junil Won, Se Jin Park, and Yong Man Ro
- arXiv Preprint, 2025

Towards Inclusive Communication: A Unified Framework for Generating Spoken Language from Sign, Lip, and Audio

- Jeonghun Yeo, **Hyeongseop Rha**, Sungjune Park, Junil Won, and Yong Man Ro
- arXiv Preprint, 2025

2024

Let's Go Real Talk: Spoken Dialogue Model for Face-to-Face Conversation

- Se Jin Park*, Chae Won Kim*, **Hyeongseop Rha**, Minsu Kim, Joanna Hong, Jeonghun Yeo, and Yong Man Ro
- Proceedings of the Annual Meeting of the Association for Computational Linguistics (**ACL Oral**), 2024

Efficient Multilingual Visual Speech Recognition by Modeling Discretized Visual Speech Units

- Minsu Kim*, Jeonghun Yeo*, Se Jin Park, **Hyeongseop Rha**, and Yong Man Ro
- The Association for Computing Machinery's Annual Conference on Multimedia (**ACMMM**), 2024

TMT: Tri-Modal Translation between Speech, Image, and Text by Processing Different Modalities as Different Languages

- Minsu Kim, Jee-weon Jung, **Hyeongseop Rha**, Soumi Maiti, Siddhant Arora, Xuankai Chang, Shinji Watanabe, and Yong Man Ro
- IEEE Transactions on Multimedia (**IEEE TMM**), Accepted 2026 (Submitted 2024)

AV-EmoDialog: Chat with Audio-Visual Users Leveraging Emotional Cues

- Se Jin Park, Yeonju Kim, **Hyeongseop Rha**, Bella Godiva, and Yong Man Ro
- arXiv Preprint, 2024

2022

Digestive Organ Recognition in Video Capsule Endoscopy Based on Temporal Segmentation Network

- Yejee Shin, Taejoon Eo, **Hyeongseop Rha**, Dong Jun Oh, Geonhui Son, Jiwoong An, You Jin Kim, Dosik Hwang, and Yun Jeong Lim
- International Conference on Medical Image Computing and Computer-Assisted Intervention (**MICCAI**), 2022

PATENTS

Neural Network-Based Medical Image Processing Apparatus and Method

- Hong Young Jeong, Yun Jeong Lim, Tae Joon Eo, Do Sik Hwang, Geon Hui Son, Ye Jee Shin, **Hyeong Seop Rha**, and Ji Woong An
- US Patent Application 18/992,843, 2026

REVIEWER ACTIVITIES

International Journal

- IEEE Transactions on Circuits and Systems for Video Technology (**TCSVT**)
- IEEE Transactions on Image Processing (**TIP**)

International Conference

- Association for the Advancement of Artificial Intelligence (**AAAI**)

SKILLS

Programming : Python, PyTorch

Languages : Korean (Native), English

AWARDS & HONORS

KAIST Fellowship

2024 – Present

National Government Fellowship

2023 – 2024